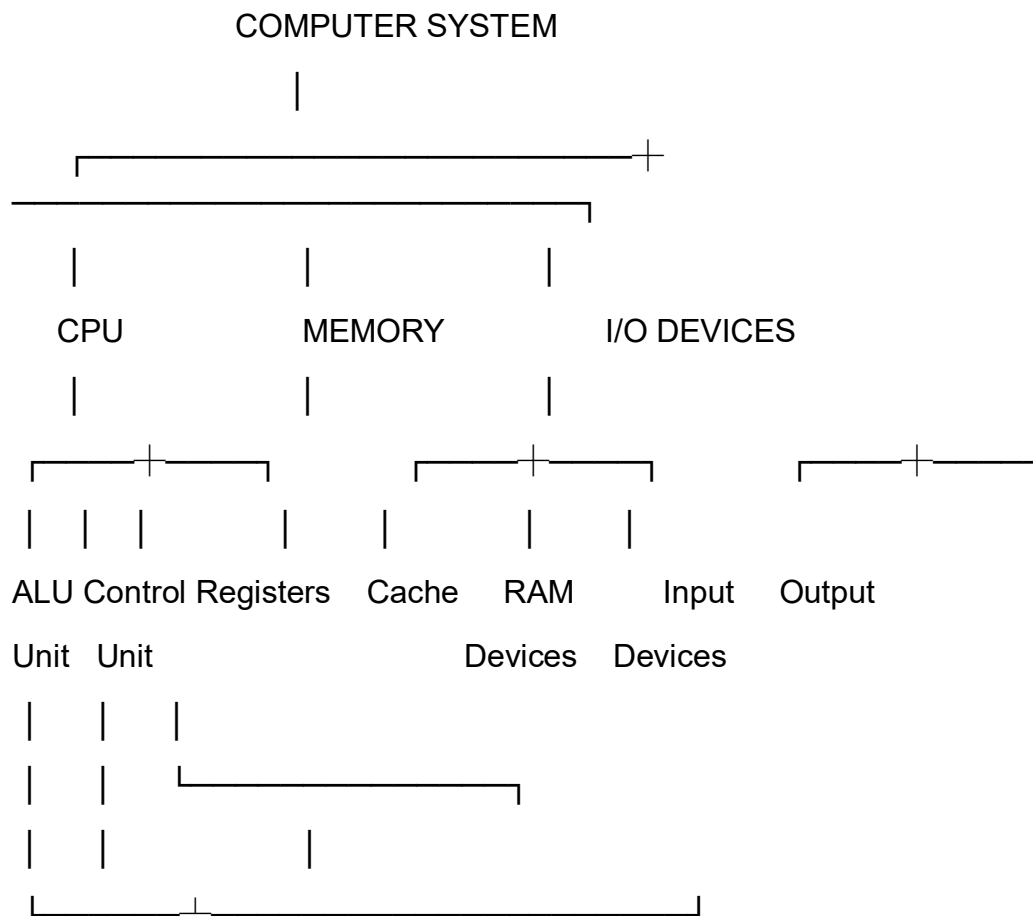
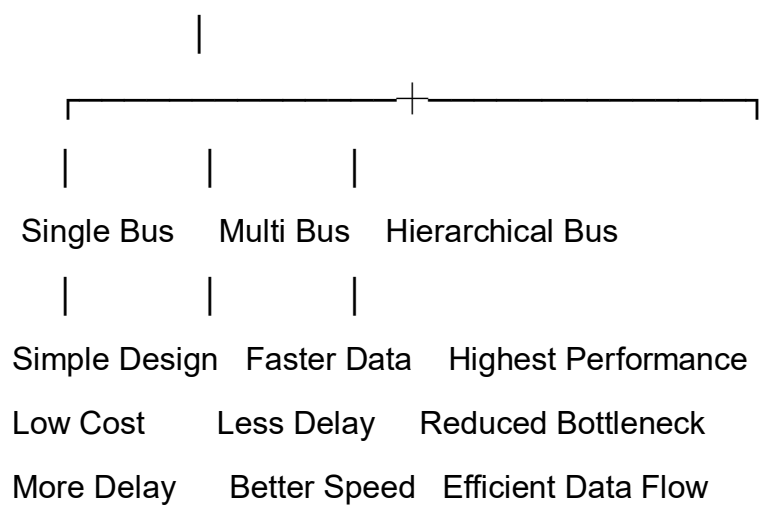


Q1. CPU, Memory, I/O Devices & Bus Structures



communicates through

BUS



Data Flow:

Input → CPU → Memory ↔ CPU → Output

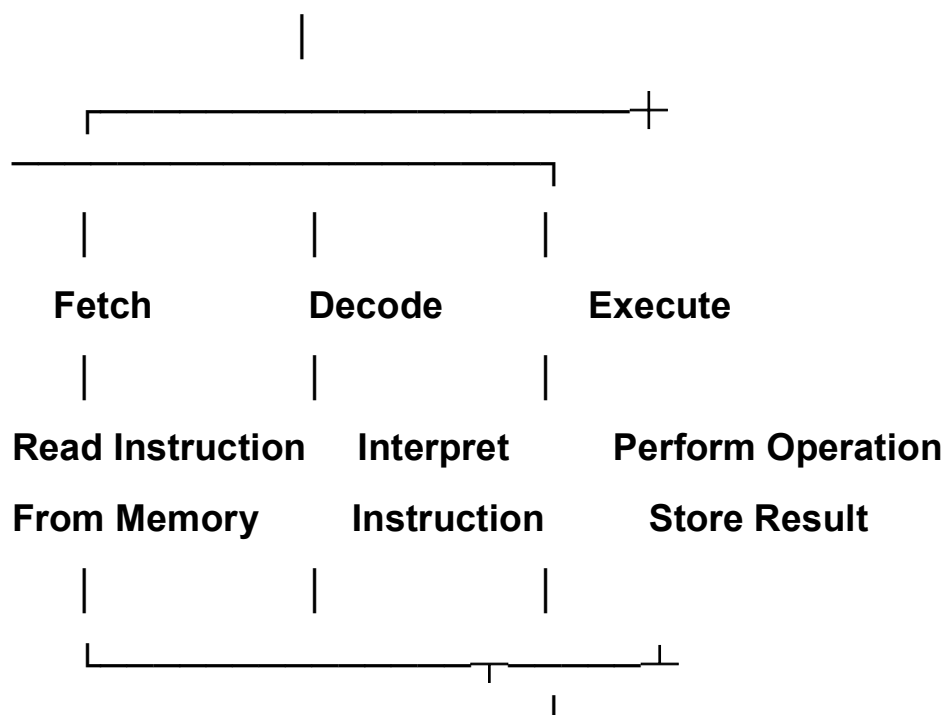
Control Signals:

Control Unit → Memory

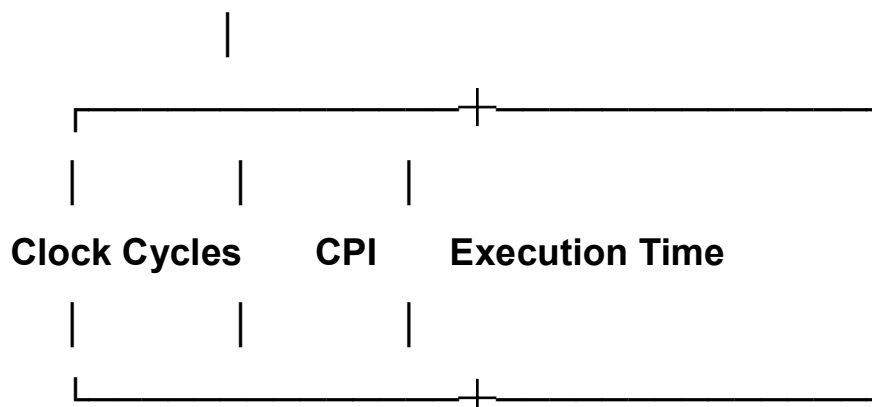
Control Unit → I/O Devices

Q2. Instruction Execution Cycle & Performance

INSTRUCTION EXECUTION CYCLE

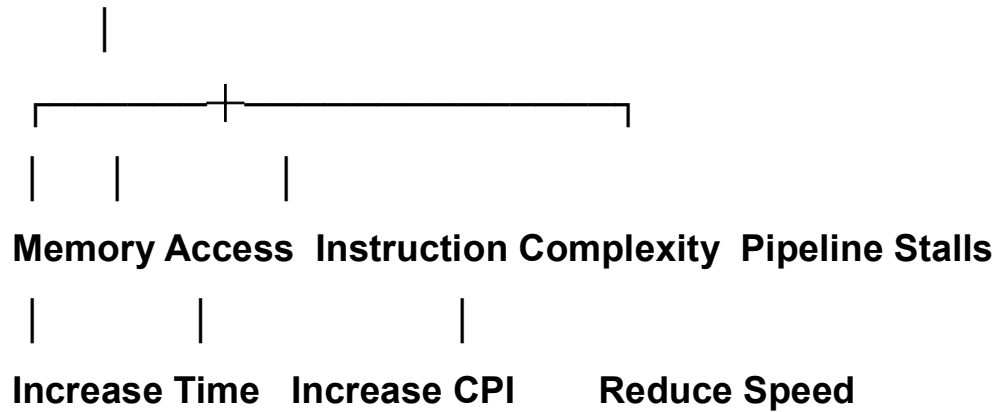


CPU PERFORMANCE



$$\text{Execution Time} = \text{CPI} \times \text{Clock Cycles} \times \text{Clock Period}$$

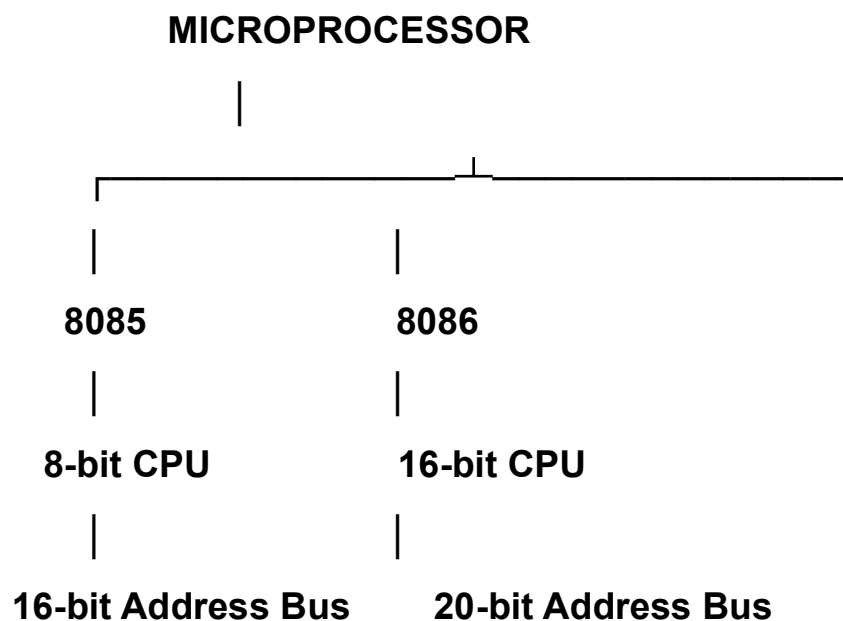
Factors Affecting Performance

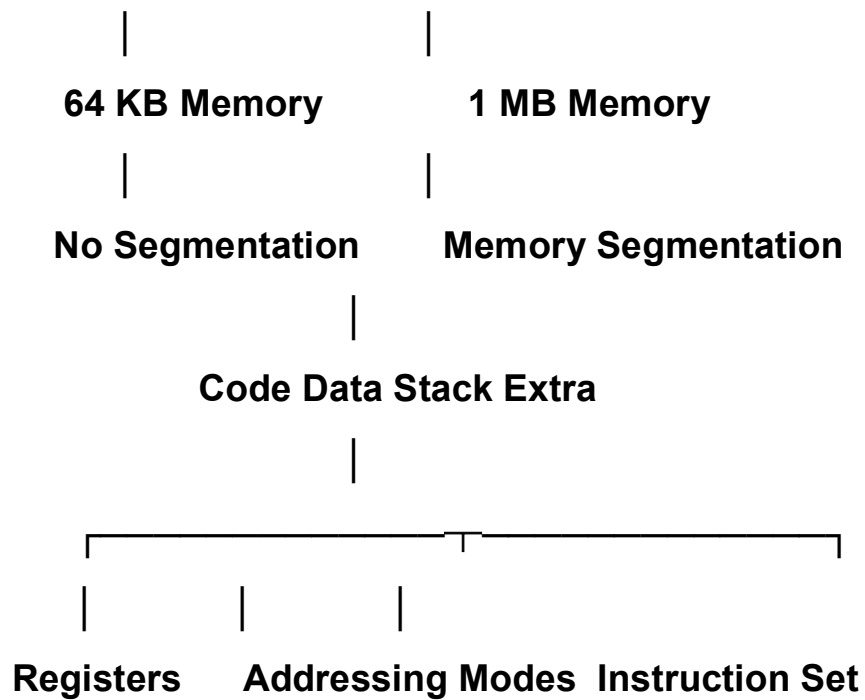


Benchmarking

Measures CPU Performance
Compares Different Processors

Q3. 8085 vs 8086 Architecture





Accumulator Immediate Larger
 General Reg. Direct Faster
 Indirect More Powerful

Effects

8085 → Simple Programs

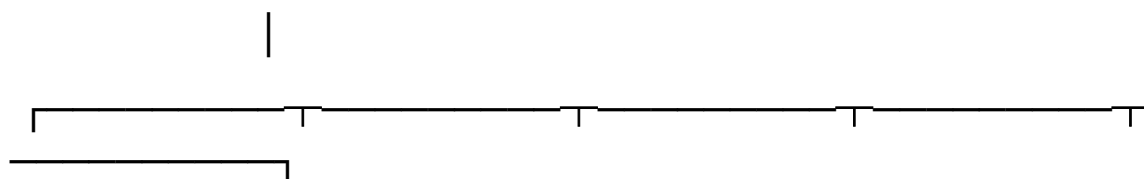
8086 → Faster Execution

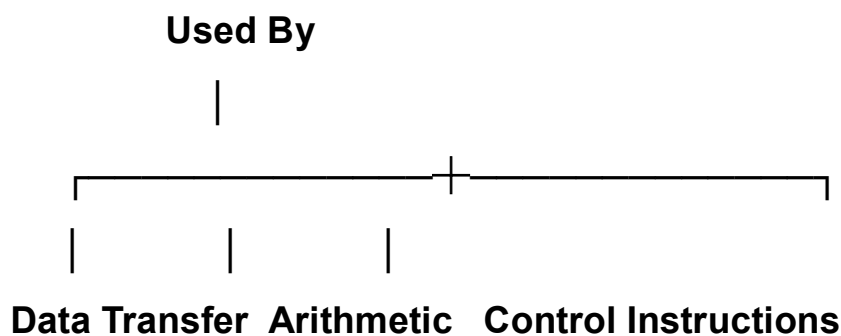
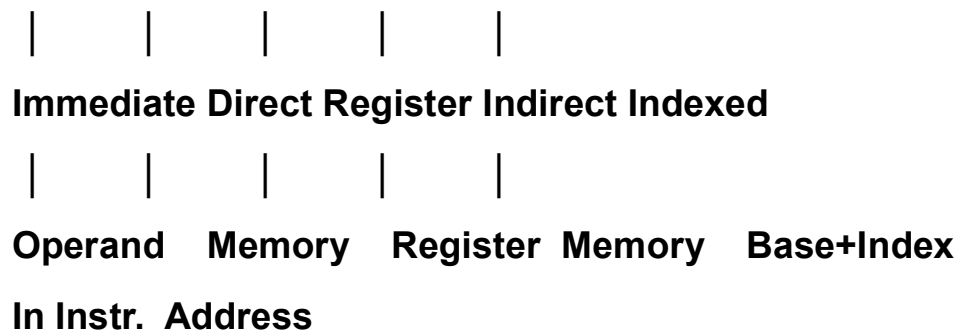
Better Multitasking

Higher Efficiency

Q4. Addressing Modes & Instruction Types

ADDRESSING MODES





Effects

- |
- |— Immediate → Fastest Execution
- |— Register → Fast Memory Access
- |— Direct → Simple but Slower
- |— Indirect → Flexible
- |— Indexed → Arrays & Loops

Real Applications

Indexed → Arrays

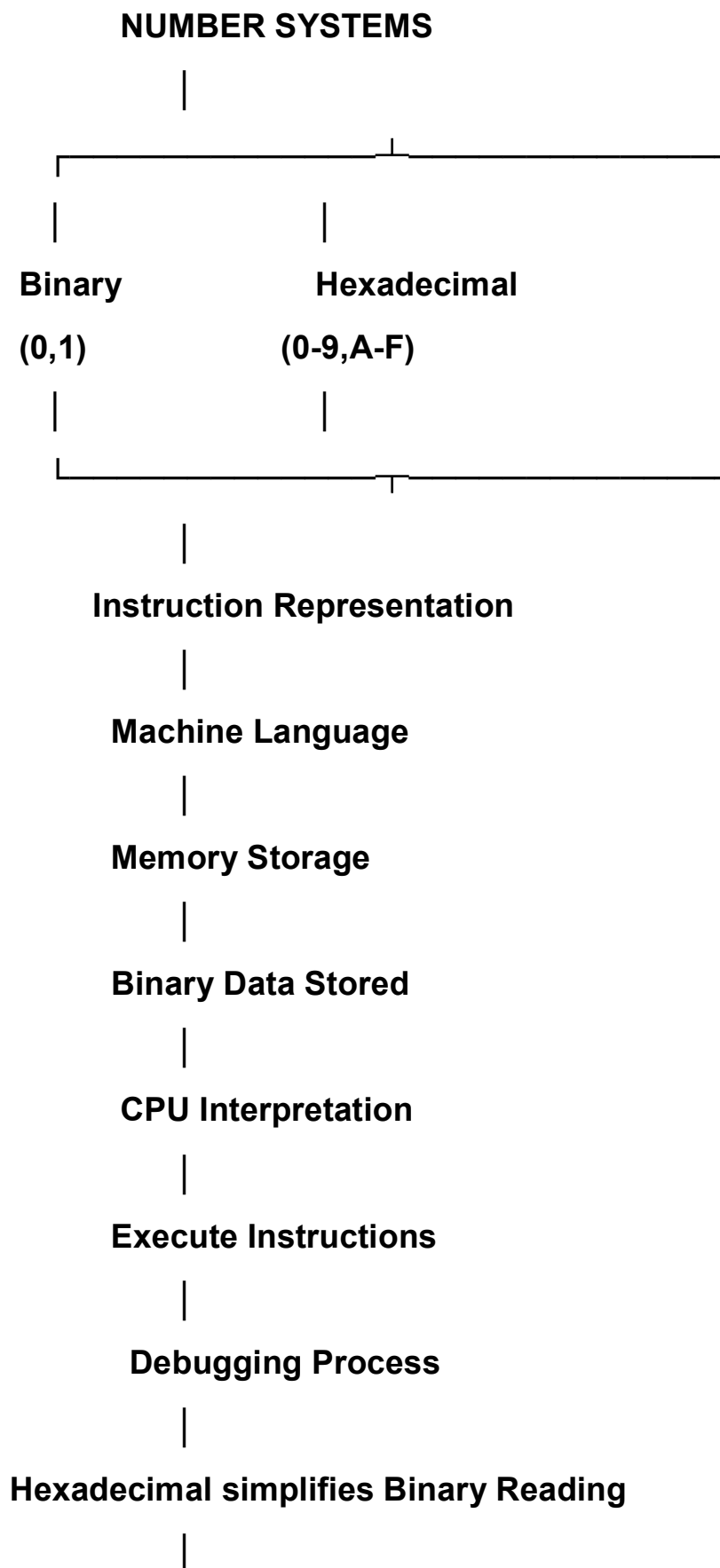
Register → Calculations

Direct → Fixed Memory

Indirect → Pointers

Immediate → Constants

Q5. Number Systems & Instruction Representation



Easier Memory Address Viewing

Conversion

Binary $\rightleftharpoons$ Hexadecimal

Used in:

Memory Addresses

Machine Code

Debugging

Low-Level Programming

System Design